

Assumptions register

Every number that appears anywhere in this submission falls into exactly one of three categories. This file lists all of them.

CATEGORY	WHAT IT MEANS	WHERE IT COMES FROM
Measured	Computed by the prototype from the simulated corpus. Reproducible by running the repo.	<code>outputs/metrics.md</code> , <code>outputs/before_after.md</code>
Simulated	A property of the corpus we generated, not of any real deployment.	<code>data/generate.py</code>
Assumed	An input we chose. Stated here with its formula so a reader can substitute their own.	this file

Nothing in this submission is a measurement of a real enterprise system. We have no access to one, and the brief does not require it.

1. Measured

Produced by `python scripts/run_eval.py` and `python scripts/run_demo.py`.

- Precision, recall, false-positive rate and false-negative rate per risk family, on the generated corpus and on the hand-written holdout
- Fast Lane and Slow Lane latency (p50 / p95 / p99), detector compute only
- Action mix under each of the three policy packs
- Threshold sweeps: what each sensitivity setting catches and what it costs
- Defect occurrence counts and clustering
- Escape reduction and the false-positive cost of the fix in the demo loop

Caveat that travels with all of these: they are measured on 264 generated traces plus 30 hand-written ones. They characterise the detectors against *this* corpus. They are not an estimate of production performance, and we do not present them as one.

2. Simulated

Properties we chose when building the corpus:

PROPERTY	VALUE	WHY
Corpus size	264 generated + 30 hand-written	Large enough for stable rates, small enough to hand-inspect
Clean traces	116 of 264 (44%)	So precision means something
Hard negatives	~30% of clean traces	Responses that look risky and are not: correct figures written as words, the user's own contact details echoed back, compliant refusals containing words like "waive", order identifiers shaped like phone numbers
Seed	20260902, fixed	Regenerating gives byte-identical traces
Identifiers	Aadhaar with valid Verhoeff checksum, cards with valid Luhn	Exercises the deterministic checks realistically. All are fabricated and correspond to no real person
Counterfactuals	Pre-recorded on each decision-support trace	In production these come from a shadow call to the same model; pre-recording makes the bias screen reproducible offline

3. Assumed

These are inputs, not findings. Each lives in the `economics:` block of a use case's contract, so changing one number changes every rupee figure in the app and the evidence pack.

Volume

ASSUMPTION	VALUE	BASIS
Customer support interactions	12,000 / week	The brief's "tens of thousands of interactions per week across these use cases combined"
Internal knowledge interactions	9,000 / week	Same
Decision support interactions	2,500 / week	Same; lower volume, higher stakes
Weeks per month	4.33	52 / 12

Cost per occurrence

ASSUMPTION	VALUE	BASIS
Rework cost (customer support)	Rs 14	Extra tokens for the retry turn plus the re-ask, at prevailing frontier-model API pricing
Rework cost (internal)	Rs 9	Shorter turns
Rework cost (decision support)	Rs 22	Longer context
Handoff rate (customer support)	7%	Share of failed interactions that reach a human agent
Handoff cost (customer support)	Rs 95	One agent contact
Handoff cost (internal)	Rs 240	An employee's own time, which is more expensive than an agent's
Handoff cost (decision support)	Rs 420	A licensed reviewer's time
Handoff rate (decision support)	100%	Every flagged case goes to a human by design
Churn exposure (customer support)	Rs 28	Expected-value placeholder, not an observed churn cost

The formula

For one defect:

occurrence_rate	= occurrences in corpus / corpus size for that use case
monthly_occurrences	= occurrence_rate x interactions_per_week x 4.33
cost_per_occurrence	= rework + (handoff_rate x handoff_cost) + churn_exposure
monthly_cost	= monthly_occurrences x cost_per_occurrence

The prototype prints this formula with its inputs next to every rupee figure it produces. See `controlplane/defects.py:price`.

What this figure is not. It is an order-of-magnitude sizing of one failure mode under stated assumptions. It is not a measured loss, and no part of it has been validated against a real cost base. A reader who disagrees with the handoff cost should change it in the contract and re-read the number.

4. Round 1 figures we withdrew

The Round 1 deck carried figures that read as measurements of a real deployment. They were illustrative, and presenting them without that label would not survive a serious question. Their status now:

ROUND 1 FIGURE	STATUS
Rs 2,14,000 / month for one defect	Replaced. The prototype now prices each defect from its own corpus with the formula shown.
4,412 occurrences / 30 days	Replaced by measured occurrence counts.
-94% occurrence reduction	Withdrawn. The measured escape reduction in the demo loop is 50%, on a 30-trace regression set, and it costs 11 additional false positives. That is the number we now use.
6.2x necessary spend	Withdrawn. Not measured; the behaviour detector counts redundant tool calls but does not price them.
25-45% of spend recovered	Withdrawn. No basis.
<100 ms p95 inline	Restated. Fast Lane detector compute is measured and reported in <code>outputs/latency.json</code> . It is compute only, excluding network and serialisation, and is stated that way.